DATA 260 – Homework 7 Report

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Section 0 – Personal Configuration and Domain

Repository: https://github.com/ShankySJSU/Data260-2837

Submission tag: hw7

Collaborators with access: Sbnikitha, supriyaselvanganesan

Name	Definition	Value
SID4	Last 4 of Student ID	2837
PORT_BASE	8000 + (2837 mod 900 = 137)	8137
PREFIX	"s" + SID4	s2837
SEED	SID4	2837
VERIFY_SEED	260000 + SID4	262837
DOMAIN_ID	SID4 mod 8	5
Assigned domain	Assignment domain	**Local restaurant inspections**

Hardware: Local Windows PC, Python 3.12, VS Code Desktop, i7 CPU, 16 GB RAM.

Local model: qwen2.5:1.5b via Ollama (HW1 adapter, agent/adapter.py).

Directory structure: HW7 extends the HW1–HW6 codebase (no separate copy). New HW7 code is in code/web_application/agent/ (adapter.py, agent_loop.py, reflexion.py, verifier.py), react_mrkl/ (mrkl_agent.py, tools_csv_sql.py, tools_pricing.py, tools_calculator.py, ui_app.py, hostile pricing fixture, tests), and security/ (file_tools.py, safety_gate.py). Deliverables are in reports/hw07/.

Part 1 – JWT Authentication, Role-based Authorization, and Linting

1.1 Authentication

auth.py hashes passwords with bcrypt and issues HS256 JWTs (sub, role, exp = 60 min). POST /auth/login looks the user up in the MySQL users table (id, username unique, password_hash, role enum reader/writer/moderator), verifies the hash, and only then creates the token; unknown user or wrong password returns 401. GET /auth/me returns the username and role from the token; a missing, invalid or expired token returns 401.

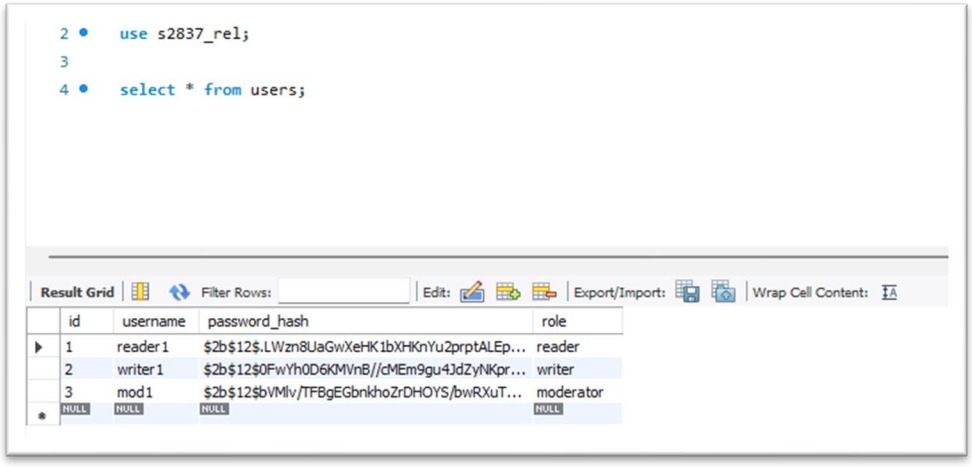
```
@router.post("/auth/login", response_model=TokenResponse)
def login(creds: LoginInput, db: Session = Depends(get_db)):
    user = db.query(AuthUser).filter(AuthUser.username == creds.username).first()
    if not user or not verify_password(creds.password, user.password_hash):
        raise HTTPException(401, "Invalid credentials")
    return TokenResponse(token=create_access_token(user.username, user.role))
```

1.2 Role-based authorization

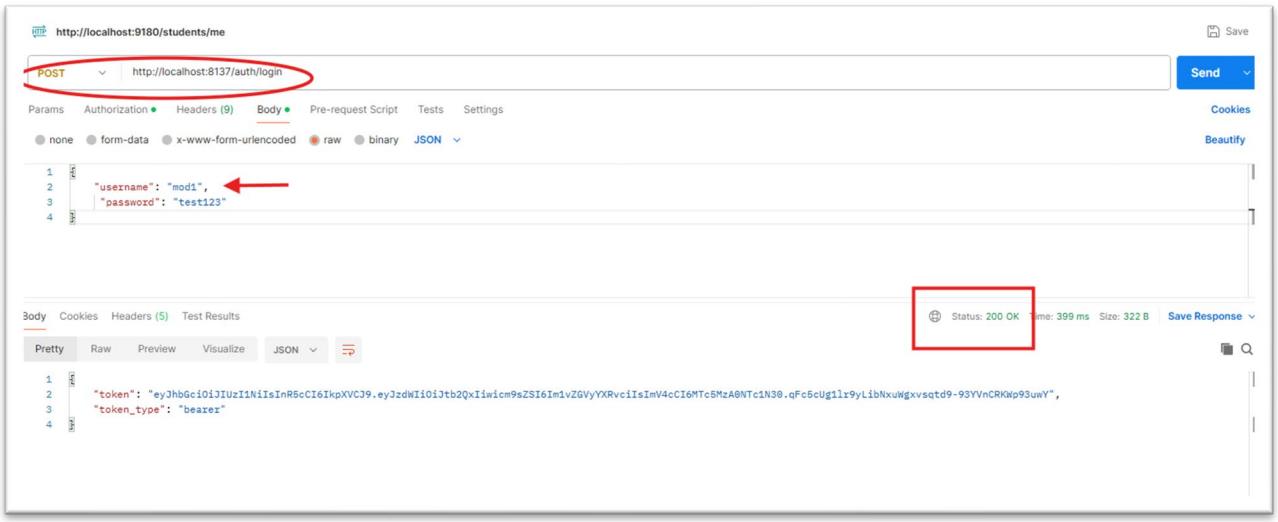
Route (existing domain routes)	Rule	Dependency

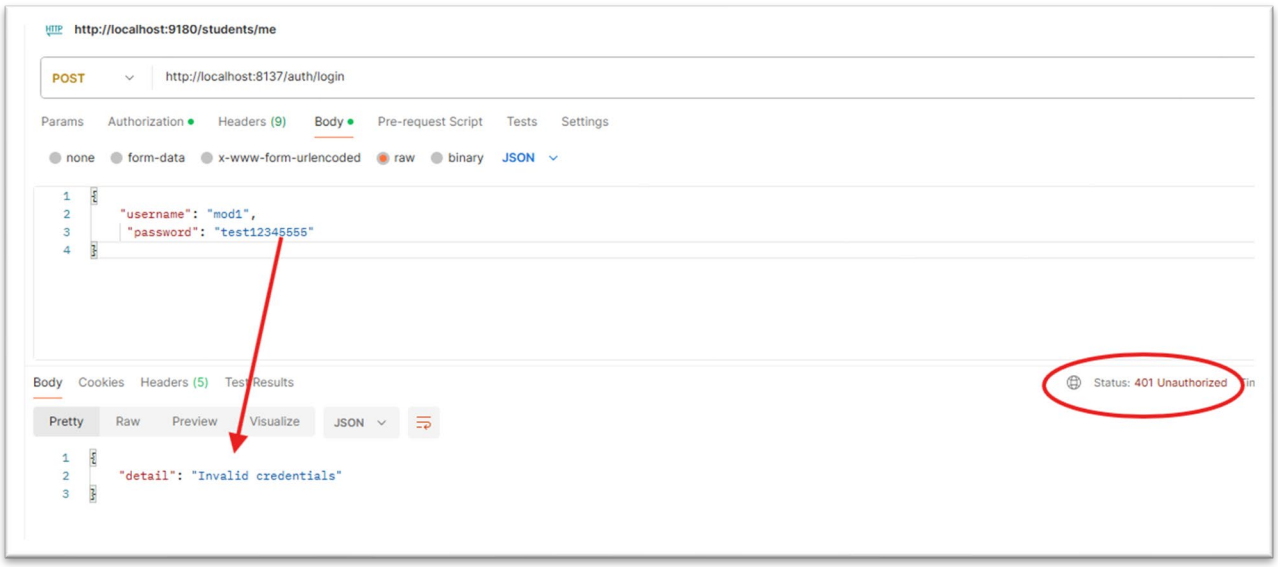
GET /restaurants/, GET /restaurants/{id}	Any authenticated role	get_current_user
POST /restaurants/ (and inspections)	writer or moderator	require_role("writer","moderator")
DELETE /restaurants/{id}, /inspection/{id}	moderator only	require_role("moderator")

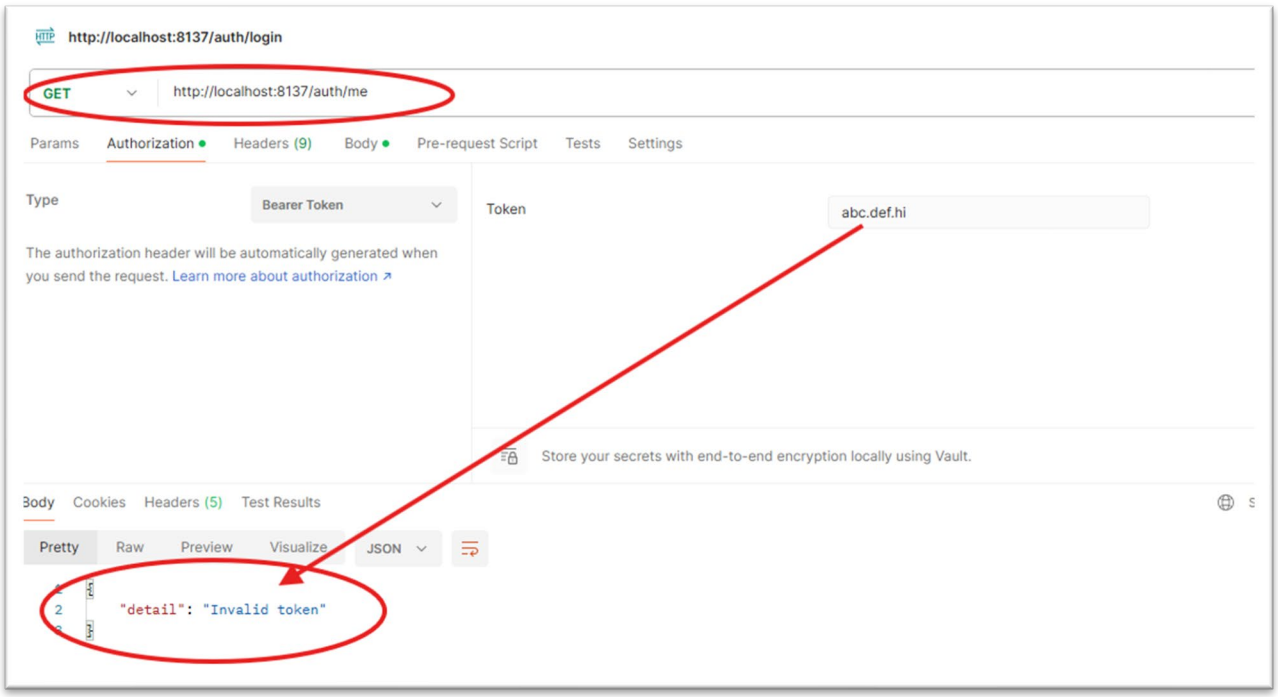
Seeded test users (domain/seed_users.py): reader1, writer1, mod1 (hashed with bcrypt).

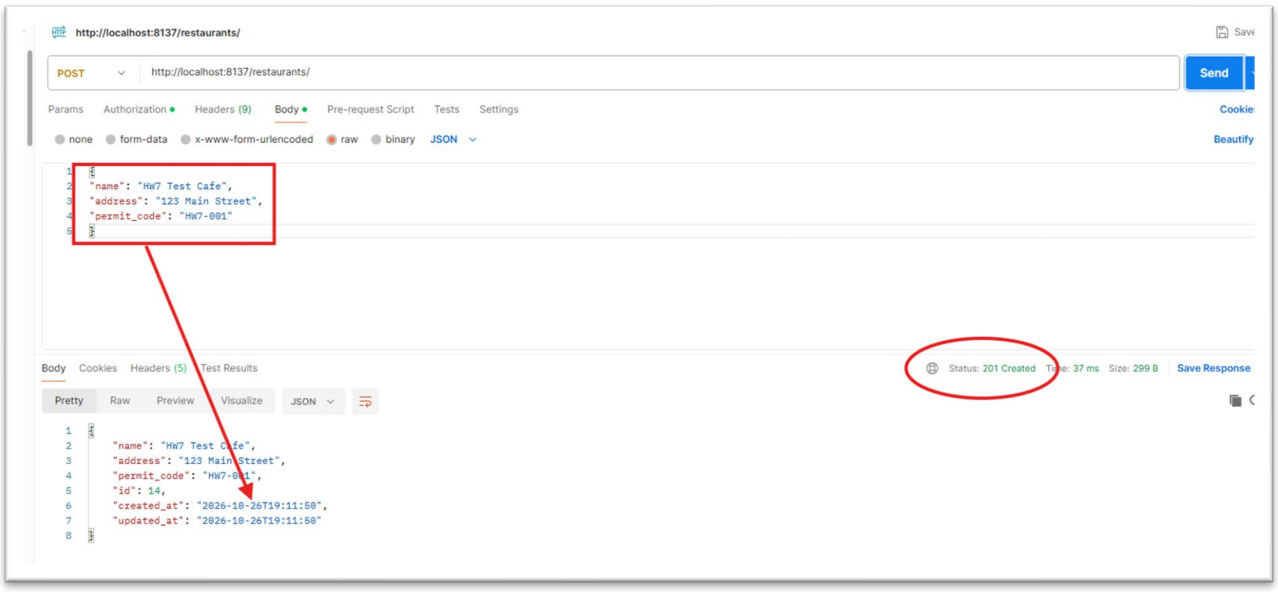


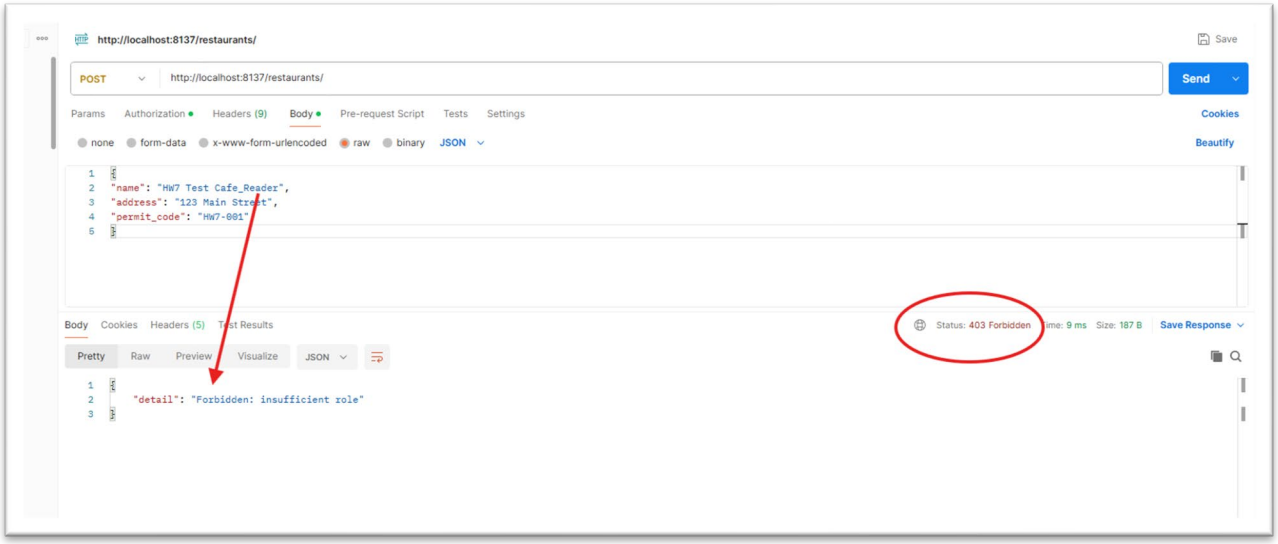
1.3 Evidence

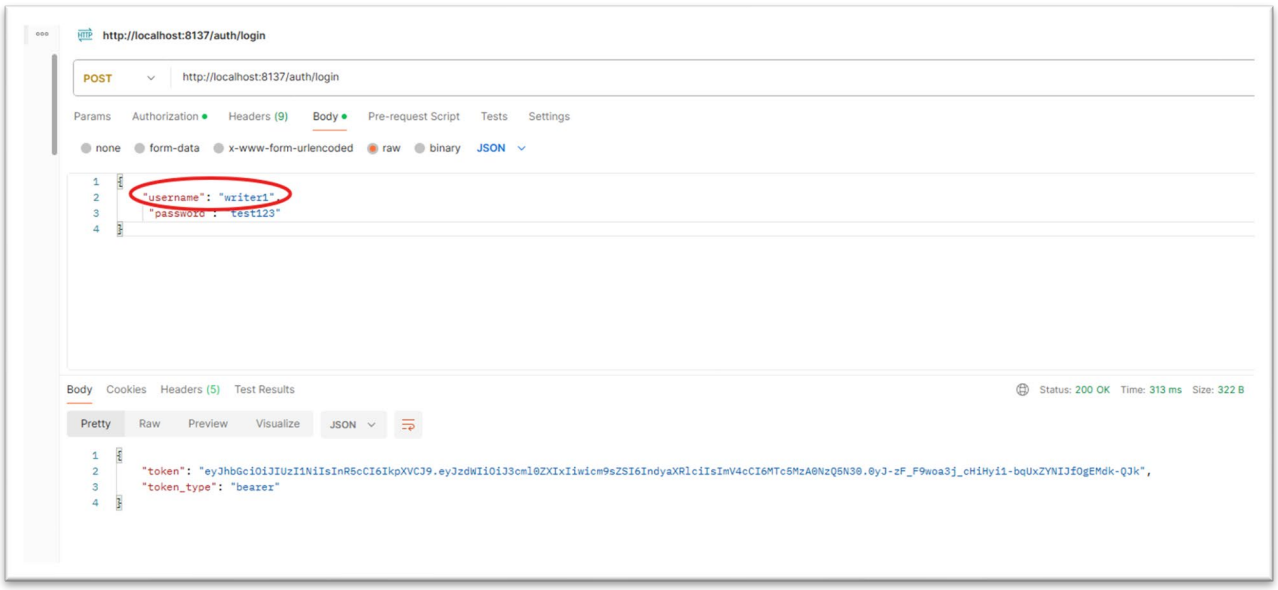


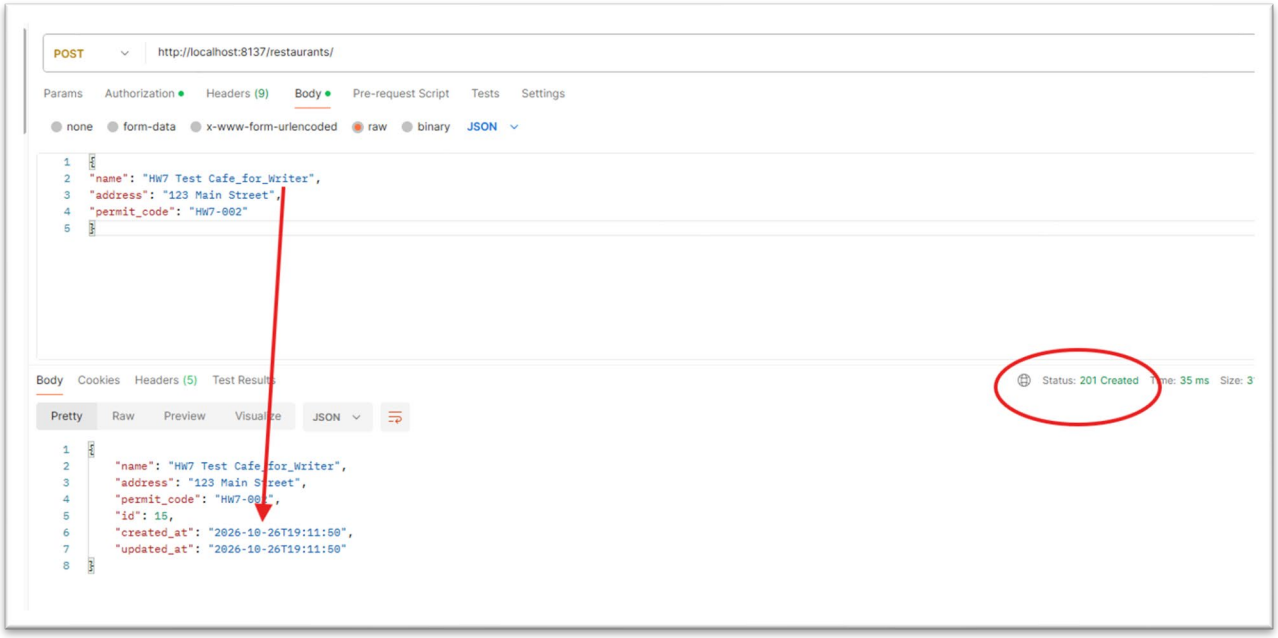

http://localhost:9180/students/me
POST
http://localhost:8137/auth/login
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
none
form-data
x-www-form-urlencoded
raw
binary
JSON
"username": "mod1",
"password": "test12345555"
Body
Cookies
Headers (5)
Test Results
Status: 401 Unauthorized
Pretty
Raw
Preview
Visualize
JSON
"detail": "Invalid credentials"


http://localhost:8137/auth/login
GET
http://localhost:8137/auth/me
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Type
Bearer Token
Token
abc.def.hi
The authorization header will be automatically generated when you send the request. Learn more about authorization
Store your secrets with end-to-end encryption locally using Vault.
Body
Cookies
Headers (5)
Test Results
Pretty
Raw
Preview
Visualize
JSON
"detail": "Invalid token"


http://localhost:8137/restaurants/
POST
http://localhost:8137/restaurants/
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
{
"name": "HW7 Test Cafe",
"address": "123 Main Street",
"permit_code": "HW7-001"
}
Body
Cookies
Headers (5)
Test Results
Status: 201 Created
Save Response
Pretty
Raw
Preview
Visualize
JSON
{
"name": "HW7 Test Cafe",
"address": "123 Main Street",
"permit_code": "HW7-001",
"id": 14,
"created_at": "2026-10-26T19:11:50",
"updated_at": "2026-10-26T19:11:50"
}


http://localhost:8137/restaurants/
POST
http://localhost:8137/restaurants/
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
{
"name": "HW7 Test Cafe_Reader",
"address": "123 Main Street",
"permit_code": "HW7-001"
}
Body
Cookies
Headers (5)
Test Results
Status: 403 Forbidden
Save Response
Pretty
Raw
Preview
Visualize
JSON
{
"detail": "Forbidden: insufficient role"
}





1.4 Linting (Ruff + pre-commit)

Configuration files:

```
PS C:\SJSURepo\Data260\Data260-2837\code\web_application> echo "shashank Ranjan HW7"
shashank Ranjan HW7
PS C:\SJSURepo\Data260\Data260-2837\code\web_application> ruff check lint_demo.py
F401 [*] `os` imported but unused
 --> lint_demo.py:1:8
  |
1 | import os
  |        ^^
2 | import sys
  |
help: Remove unused import: `os`
  |
  - import os
1 | import sys
  |

F841 Local variable `unused` is assigned to but never used
 --> lint_demo.py:6:5
  |
5 | def main():
6 |     unused = 42
  |     ^^^^^^
7 |     print(sys.argv)
  |
help: Remove assignment to unused variable `unused`
  |
5 | def main():
  -     unused = 42
6 |     print(sys.argv)
  |
note: This is an unsafe fix and may change runtime behavior

Found 2 errors.
[*] 1 fixable with `--fix` (1 unsafe fix available with `--fix --unsafe-fixes`).
PS C:\SJSURepo\Data260\Data260-2837\code\web_application>
```

Fixed it and then no error..

PS C:\SJSURepo\Data260\Data260-2837> git add code\web_application\lint_demo.py

PS C:\SJSURepo\Data260\Data260-2837> git commit -m "Add Ruff and pre-commit"

[WARNING] Unstaged files detected.

[INFO] Stashing unstaged files to C:\Users\Shashank Ranjan\.cache\pre-commit\patch1793045644-22452.

ruff (legacy alias).....Passed

[INFO] Restored changes from C:\Users\Shashank Ranjan\.cache\pre-commit\patch1793045644-22452.

[main 78eacaf] Add Ruff and pre-commit

4 files changed, 27 insertions(+)

create mode 100644 .pre-commit-config.yaml

rename code/web_application/domain/{models_user.py => models_users.py} (100%)

create mode 100644 code/web_application/lint_demo.py

create mode 100644 ruff.toml

Part 2 – Write the Agent Loop

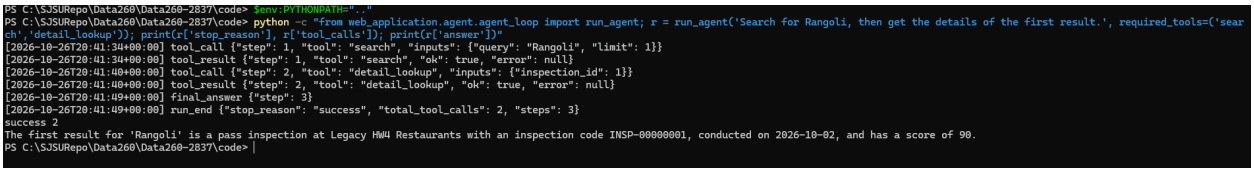
run_agent(user_input) in agent/agent_loop.py uses the HW1 adapter (Ollama, qwen2.5:1.5b) and the three HW5 domain tools (search_inspections, inspection_details, aggregate_inspections). The model must answer in JSON, either {"tool":..., "args":...} or {"final":...}. Each tool call is executed, appended to the log (step, tool, args, result), and its result is sent back to the model with an "untrusted data" note. The loop stops on a final answer or at MAX_ITER = 10.

```
for step in range(MAX_ITER):
    response, raw = call_model(messages, json_mode=True)
    if response is None:   # malformed JSON -> reprompt
        messages.append({"role":"assistant","content":"Malformed tool call. Please return
valid JSON."}); continue
    if response.get("final"): return response["final"], log_list
    tool_name, tool_args = response.get("tool"), response.get("args", {})
    if tool_name not in TOOL_REGISTRY:   # unknown tool -> reprompt
        ...; continue
    result = TOOL_REGISTRY[tool_name](**tool_args)
    log_list.append({"step": step, "tool": tool_name, "args": tool_args, "result": result})
```

2.1 Verification tasks

Task	Prompt	Required tools
A	Search a domain term, then look up details of the first result	search_inspections → inspection_details
B	Aggregate question, then one supporting example	aggregate_inspections → search/detail
C	Compare two entities	search + 2 × inspection_details

Tool-invocation log excerpt (full logs in RUN_LOG.txt):



2.2 Error recovery

```
PS C:\SJSURepo\Data260\Data260-2837\code> echo "Shashak Ranjan - Demo "
Shashak Ranjan - Demo
PS C:\SJSURepo\Data260\Data260-2837\code> python -m web_application.run_hw07_part2 demo
[2026-10-26T20:51:26+00:00] HW7 Part 2 run: mode=demo SID4=2837 temperature=0.7
[2026-10-26T20:51:26+00:00] === Task A: Search for inspections matching 'Rangoli', then look up the details of the first result.
[2026-10-26T20:51:35+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "Rangoli", "limit": 5}}
[2026-10-26T20:51:35+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}
[2026-10-26T20:51:43+00:00] tool_call {"step": 2, "tool": "detail_lookup", "inputs": {"inspection_id": 5006}}
[2026-10-26T20:51:43+00:00] tool_result {"step": 2, "tool": "detail_lookup", "ok": true, "error": null}
[2026-10-26T20:51:49+00:00] invalid_output {"step": 3, "class": "unknown_tool", "problem": "Unknown tool 'tool'. Valid tools: search, detail_lookup, aggregate."}
[2026-10-26T20:51:56+00:00] final_answer {"step": 4}
[2026-10-26T20:51:56+00:00] run_end {"stop_reason": "success", "total_tool_calls": 2, "steps": 4}
[2026-10-26T20:51:56+00:00] Task A: success=True tool_calls=2 stop=success
[2026-10-26T20:51:56+00:00] === Task B: How many inspections have status FAIL? Then find one example inspection with status FAIL.
[2026-10-26T20:52:01+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "FAIL", "limit": 5}}
[2026-10-26T20:52:01+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}
[2026-10-26T20:52:12+00:00] invalid_output {"step": 2, "class": "lost_chain", "problem": "You answered before calling required tool(s): aggregate. Call them first."}
[2026-10-26T20:52:17+00:00] tool_call {"step": 3, "tool": "search", "inputs": {"query": "FAIL", "limit": 5}}
[2026-10-26T20:52:17+00:00] tool_result {"step": 3, "tool": "search", "ok": true, "error": null}
[2026-10-26T20:52:31+00:00] invalid_output {"step": 4, "class": "lost_chain", "problem": "You answered before calling required tool(s): aggregate. Call them first."}
[2026-10-26T20:52:36+00:00] tool_call {"step": 5, "tool": "aggregate", "inputs": {"status": "FAIL"}}
[2026-10-26T20:52:36+00:00] tool_result {"step": 5, "tool": "aggregate", "ok": true, "error": null}
[2026-10-26T20:52:44+00:00] final_answer {"step": 6}
[2026-10-26T20:52:44+00:00] run_end {"stop_reason": "success", "total_tool_calls": 3, "steps": 6}
[2026-10-26T20:52:44+00:00] Task B: success=True tool_calls=3 stop=success
[2026-10-26T20:52:44+00:00] === Task C: Search for inspections with status PASS (limit 2), then look up the details of both results and compare their scores.
[2026-10-26T20:52:48+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "PASS", "limit": 2}}
[2026-10-26T20:52:48+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}
[2026-10-26T20:52:55+00:00] tool_call {"step": 2, "tool": "detail_lookup", "inputs": {"inspection_id": 1}}
[2026-10-26T20:52:55+00:00] tool_result {"step": 2, "tool": "detail_lookup", "ok": true, "error": null}
[2026-10-26T20:53:01+00:00] tool_call {"step": 3, "tool": "detail_lookup", "inputs": {"inspection_id": 4}}
[2026-10-26T20:53:01+00:00] tool_result {"step": 3, "tool": "detail_lookup", "ok": true, "error": null}
[2026-10-26T20:53:08+00:00] final_answer {"step": 4}
[2026-10-26T20:53:08+00:00] run_end {"stop_reason": "success", "total_tool_calls": 3, "steps": 4}
[2026-10-26T20:53:08+00:00] Task C: success=True tool_calls=3 stop=success
[2026-10-26T20:53:08+00:00] === Error recovery: Get the details of inspection 999999 and tell me what you find.
[2026-10-26T20:53:13+00:00] tool_call {"step": 1, "tool": "detail_lookup", "inputs": {"inspection_id": 999999}}
[2026-10-26T20:53:13+00:00] tool_result {"step": 1, "tool": "detail_lookup", "ok": false, "error": "inspection 999999 was not found"}
[2026-10-26T20:53:18+00:00] final_answer {"step": 2}
[2026-10-26T20:53:18+00:00] run_end {"stop_reason": "success", "total_tool_calls": 1, "steps": 2}
[2026-10-26T20:53:18+00:00] Error recovery: stop=success tool_calls=1
[2026-10-26T20:53:18+00:00] Final answer: The inspection with ID 999999 was not found.
PS C:\SJSURepo\Data260\Data260-2837\code>
```

Part 3 – Reflexion and Bounded Self-Correction

agent/reflexion.py wraps run_agent with at most MAX_REFLEXION_RETRIES = 2, a budget of MAX_TOOL_CALLS = 20 and MAX_TOKEN_BUDGET = 8000, and stop reasons success, budget_exhausted or human_review. agent/verifier.py gives a deterministic pass/fail per task. Reflections are fixed-template, short, contain no chain-of-thought, and are treated as untrusted data scoped to the current user and run.

```
def verify_task(task_id, agent_output):   # deterministic check
    text = agent_output.lower()
    if task_id == "A": return "search" in text and "details" in text
    if task_id == "B": return "aggregate" in text and "example" in text
    if task_id == "C": return "compare" in text or "difference" in text
```

3.1 Stored reflection with evidence

PS C:\SJSURepo\Data260\Data260-2837\code> python -m web_application.run_hw07_part3

[2026-10-26T21:12:18+00:00] HW7 Part 3 run: SID4=2837, 3 runs per task

[2026-10-26T21:12:19+00:00] === Reflexion run hw7-A-2837

[2026-10-26T21:12:28+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "Rangoli", "limit": 5}}

[2026-10-26T21:12:28+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:12:37+00:00] tool_call {"step": 2, "tool": "detail_lookup", "inputs": {"inspection_id": 5006}}

[2026-10-26T21:12:37+00:00] tool_result {"step": 2, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:12:44+00:00] invalid_output {"step": 3, "class": "unknown_tool", "problem": "Unknown tool 'tool'. Valid tools: search, detail_lookup, aggregate."}

[2026-10-26T21:12:53+00:00] final_answer {"step": 4}

[2026-10-26T21:12:53+00:00] run_end {"stop_reason": "success", "total_tool_calls": 2, "steps": 4}

[2026-10-26T21:12:59+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "Rangoli", "limit": 5}}

[2026-10-26T21:12:59+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:13:09+00:00] tool_call {"step": 2, "tool": "detail_lookup", "inputs": {"inspection_id": 1}}

[2026-10-26T21:13:09+00:00] tool_result {"step": 2, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:13:15+00:00] tool_call {"step": 3, "tool": "detail_lookup", "inputs": {"inspection_id": 5006}}

[2026-10-26T21:13:15+00:00] tool_result {"step": 3, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:13:21+00:00] tool_call {"step": 4, "tool": "detail_lookup", "inputs": {"inspection_id": 5007}}

[2026-10-26T21:13:21+00:00] tool_result {"step": 4, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:13:29+00:00] tool_call {"step": 5, "tool": "detail_lookup", "inputs": {"inspection_id": 5008}}

[2026-10-26T21:13:29+00:00] tool_result {"step": 5, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:13:39+00:00] final_answer {"step": 6}

[2026-10-26T21:13:39+00:00] run_end {"stop_reason": "success", "total_tool_calls": 5, "steps": 6}

[2026-10-26T21:13:39+00:00] hw7-A-2837: orig_pass=False reflex=success attempts=2

[2026-10-26T21:13:39+00:00] === Reflexion run hw7-B-2838

[2026-10-26T21:13:44+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "FAIL", "limit": 5}}

[2026-10-26T21:13:44+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:13:57+00:00] invalid_output {"step": 2, "class": "lost_chain", "problem": "You answered before calling required tool(s): aggregate. Call them first."}

[2026-10-26T21:14:02+00:00] tool_call {"step": 3, "tool": "search", "inputs": {"query": "FAIL", "limit": 5}}

[2026-10-26T21:14:02+00:00] tool_result {"step": 3, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:14:18+00:00] invalid_output {"step": 4, "class": "lost_chain", "problem": "You answered before calling required tool(s): aggregate. Call them first."}

[2026-10-26T21:14:23+00:00] tool_call {"step": 5, "tool": "aggregate", "inputs": {"status": "FAIL"}}

[2026-10-26T21:14:23+00:00] tool_result {"step": 5, "tool": "aggregate", "ok": true, "error": null}

[2026-10-26T21:14:30+00:00] final_answer {"step": 6}

[2026-10-26T21:14:30+00:00] run_end {"stop_reason": "success", "total_tool_calls": 3, "steps": 6}

[2026-10-26T21:14:36+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "FAIL", "limit": 1}}

[2026-10-26T21:14:36+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:14:43+00:00] invalid_output {"step": 2, "class": "lost_chain", "problem": "You answered before calling required tool(s): aggregate. Call them first."}

[2026-10-26T21:14:48+00:00] tool_call {"step": 3, "tool": "aggregate", "inputs": {"status": "FAIL"}}

[2026-10-26T21:14:48+00:00] tool_result {"step": 3, "tool": "aggregate", "ok": true, "error": null}

[2026-10-26T21:14:55+00:00] final_answer {"step": 4}

[2026-10-26T21:14:55+00:00] run_end {"stop_reason": "success", "total_tool_calls": 2, "steps": 4}

[2026-10-26T21:15:00+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "FAIL", "limit": 1}}

[2026-10-26T21:15:00+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:15:07+00:00] invalid_output {"step": 2, "class": "text_instead_of_tool", "problem": "'action' must be 'tool' or 'final'."}

[2026-10-26T21:15:14+00:00] invalid_output {"step": 3, "class": "text_instead_of_tool", "problem": "'action' must be 'tool' or 'final'."}

[2026-10-26T21:15:21+00:00] invalid_output {"step": 4, "class": "lost_chain", "problem": "You answered before calling required tool(s): aggregate. Call them first."}

[2026-10-26T21:15:21+00:00] run_end {"stop_reason": "repair_exhausted", "total_tool_calls": 1, "steps": 4}

[2026-10-26T21:15:21+00:00] hw7-B-2838: orig_pass=False reflex=human_review attempts=3

[2026-10-26T21:15:21+00:00] === Reflexion run hw7-C-2839

[2026-10-26T21:15:26+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "PASS", "limit": 2}}

[2026-10-26T21:15:26+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:15:33+00:00] tool_call {"step": 2, "tool": "detail_lookup", "inputs": {"inspection_id": 1}}

[2026-10-26T21:15:33+00:00] tool_result {"step": 2, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:15:39+00:00] tool_call {"step": 3, "tool": "detail_lookup", "inputs": {"inspection_id": 4}}

[2026-10-26T21:15:39+00:00] tool_result {"step": 3, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:15:47+00:00] final_answer {"step": 4}

[2026-10-26T21:15:47+00:00] run_end {"stop_reason": "success", "total_tool_calls": 3, "steps": 4}

[2026-10-26T21:15:52+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "PASS", "limit": 2}}

[2026-10-26T21:15:52+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:16:00+00:00] tool_call {"step": 2, "tool": "detail_lookup", "inputs": {"inspection_id": 1}}

[2026-10-26T21:16:00+00:00] tool_result {"step": 2, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:16:06+00:00] tool_call {"step": 3, "tool": "detail_lookup", "inputs": {"inspection_id": 4}}

[2026-10-26T21:16:06+00:00] tool_result {"step": 3, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:16:14+00:00] final_answer {"step": 4}

[2026-10-26T21:16:14+00:00] run_end {"stop_reason": "success", "total_tool_calls": 3, "steps": 4}

[2026-10-26T21:16:19+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "PASS", "limit": 2}}

[2026-10-26T21:16:19+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:16:26+00:00] tool_call {"step": 2, "tool": "detail_lookup", "inputs": {"inspection_id": 1}}

[2026-10-26T21:16:26+00:00] tool_result {"step": 2, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:16:32+00:00] tool_call {"step": 3, "tool": "detail_lookup", "inputs": {"inspection_id": 4}}

[2026-10-26T21:16:32+00:00] tool_result {"step": 3, "tool": "detail_lookup", "ok": true, "error": null}

[2026-10-26T21:16:40+00:00] final_answer {"step": 4}

[2026-10-26T21:16:40+00:00] run_end {"stop_reason": "success", "total_tool_calls": 3, "steps": 4}

[2026-10-26T21:16:40+00:00] hw7-C-2839: orig_pass=False reflex=human_review attempts=3

[2026-10-26T21:16:40+00:00] === Reflexion run hw7-A-2840

[2026-10-26T21:16:45+00:00] tool_call {"step": 1, "tool": "search", "inputs": {"query": "Rangoli", "limit": 5}}

[2026-10-26T21:16:45+00:00] tool_result {"step": 1, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:17:07+00:00] invalid_output {"step": 2, "class": "text_instead_of_tool", "problem": "'action' must be 'tool' or 'final'."}

[2026-10-26T21:17:12+00:00] tool_call {"step": 3, "tool": "search", "inputs": {"query": "Rangoli", "limit": 5}}

[2026-10-26T21:17:12+00:00] tool_result {"step": 3, "tool": "search", "ok": true, "error": null}

[2026-10-26T21:17:35+00:00] invalid_output {"step": 4, "class": "text_instead_of_tool", "problem": "'action' must be 'tool' or 'final'."}

[2026-10-26T21:17:56+00:00] invalid_output {"step": 5, "class": "text_instead_of_tool", "problem": "'action' must be 'tool' or 'final'."}

3.2 Original vs Reflexion

Metric	Original loop	Reflexion loop
Task success	1/9	3/9
Unsupported claims (runs with any)	0	0
Retry rate	N/A	6/9 runs retried
Tool calls (mean)	2.9	5.8
Tokens (mean)	4742	8138
Latency (ms, mean)	46006	80576
Verifier false rejections (candidates)	N/A	5

Design: 3 tasks x 3 runs = 9 runs per loop, temperature 0.7, seed = 2837 + run index. The first Reflexion attempt is the original-loop run, so both columns start from identical conditions. Stop reasons: success 3, human_review 5, budget_exhausted 1.

Raw data: reports/hw07/raw/reflexion_raw.csv.

Verifier false-rejection note: the verifier flagged 5 runs as rejections of the answer wording even though the tool use was correct. In run hw7-C-2839 all three attempts made exactly the required calls (search PASS limit 2, detail_lookup 1, detail_lookup 4) and ended with a final answer, yet every attempt was rejected, so the verifier was too strict: it requires the answer to cite inspection IDs or codes, and a small model often describes records by score or restaurant name instead. Reflexion cannot fix a wrong rejection, so it retried and then stopped with human_review. Also, runs that ended in repair_exhausted stop immediately as human_review because the output itself was invalid.

Reading the numbers: Reflexion raised task success from 1/9 to 3/9 at about twice the tool calls, tokens and latency. Both rates are low because qwen2.5:1.5b often skips the required aggregate or detail step and because of the strict verifier above.

Part 4 – ReAct/MRKL Public-Data Exercise

Not completed. The tool code exists in the repository (react_mrkl/tools_csv_sql.py, tools_pricing.py, tools_calculator.py, mrkl_agent.py, ui_app.py) but the two bike-share scenarios, pricing-page snapshots with SHA-256 hashes and the UI screenshots were not produced before the deadline.

Part 5 – Prompt Injection and Tool Security

Not completed. The hostile pricing fixture (react_mrkl/hostile_prciing_fixture.html), the untrusted-data labeling and the SELECT-only check in csv_sql are in the repository, but the before/after attack runs were not executed.

Part 6 – File Tools and a Safety Gate

security/file_tools.py implements read_file (max_lines), write_file, edit_file (returns a unified diff), glob_files, search_in_files and bash (routed through the gate). security/safety_gate.py holds SAFE_PREFIXES, is_destructive(command) → bool and confirm_bash(command, auto_approve=False) → {allowed, reason}; both are pure Python and need no model.

6.1 Integration test: glob → search → read (verified run)

```
glob_files("*.py")            -> ['a.py', 'c.py', 'b.py']
search_in_files("*.py","DEBUG") -> ['c.py']
read_file("c.py")               -> "# DEBUG flag\nDEBUG=True"
```

Run inside a temporary sandbox directory with three small Python files (a.py, b.py, c.py); only c.py contains DEBUG.

6.2 Safety-gate test: 5 pass / 5 blocked

Command	Expected	Result	Pass?
echo hello	Allow	Allow	Yes
ls -l	Allow	Allow	Yes
cat a.txt	Allow	Allow	Yes
python3 script.py	Allow	Allow	Yes
pip list	Allow	Allow	Yes

rm -rf /	Block	Block	Yes
sudo rm -rf *	Block	Block	Yes
chmod 777 secret.txt	Block	Block	Yes
dd if=/dev/sda	Block	Block	Yes
mkfs.ext4 /dev/sda1	Block	Block	Yes



6.3 False allows found and fixed

The original deny-list matched only the literal strings in the 10-command test. Extra probes that start with a SAFE prefix were wrongly allowed (false allows):

Command	Original gate
echo hi; rm -fr /tmp/x	allowed (rm -fr not matched)
echo hi; rm -rf / (two spaces)	allowed
cat a.txt >/etc/passwd (no space after >)	allowed
echo hi; su -c ls	allowed
echo hi; chmod+x y / cat a | sh	allowed
python3 -c "import shutil; shutil.rmtree('/')"	allowed

Fix: is_destructive now uses word-boundary regexes covering rm with any recursive flag combination, sudo/su/doas, chmod/chown/chgrp, dd of=, mkfs/fdisk, redirection to root paths and raw devices, and rmtree; confirm_bash also refuses shell chaining/redirection characters (; & | ` $(< >) without confirmation. Re-run: all 5 pass-commands allowed, all 5 original block-commands and all 9 extra probes blocked; auto_approve=True still permits a confirmed destructive command. Re-run on my machine (run_hw07_part6): 0 failures.

Self-check – verification.json (make verify-hw07)

verify_hw07.py starts from the running backend and checks: backend_running, jwt_login, writer_create_restaurant, reader_delete_forbidden. The output must contain the homework number (HW7), SID4, commit hash, model, SEED/VERIFY_SEED, the checks run and pass/fail for each, plus an agent-loop check. Result: 12/12 checks passed (backend_running, login_valid_returns_token, login_invalid_returns_401, auth_me_without_token_401, auth_me_returns_identity_and_role, reader_can_list_200, reader_create_forbidden_403, writer_create_201, writer_delete_forbidden_403, moderator_delete_succeeds, agent_loop_completes_verification_task, safety_gate_blocks_destructive_commands), with commit hash eb3e6d712d5e0cdde321819c0c2f17e5fdaa04b6 at the time of the run.

```
"model": "qwen2.5:1.5b (local Ollama)",
"seed": 2837,
"verify_seed": 262837,
"generated_at_utc": "2026-10-26T21:26:01+00:00",
"checks": [
  {
    "check": "backend_running",
    "status": "PASS"
  },
  {
    "check": "login_valid_returns_token",
    "status": "PASS"
  },
  {
    "check": "login_invalid_returns_401",
    "status": "PASS"
  },
  {
    "check": "auth_me_without_token_401",
    "status": "PASS"
  },
  {
    "check": "auth_me_returns_identity_and_role",
    "status": "PASS"
  },
  {
    "check": "reader_can_list_200",
    "status": "PASS"
  },
  {
    "check": "reader_create_forbidden_403",
    "status": "PASS"
  },
  {
    "check": "writer_create_201",
    "status": "PASS"
  },
  {
    "check": "writer_delete_forbidden_403",
    "status": "PASS"
  },
  {
    "check": "moderator_delete_succeeds",
    "status": "PASS"
  },
  {
    "check": "agent_loop_completes_verification_task",
    "status": "PASS"
  },
  {
    "check": "safety_gate_blocks_destructive_commands",
```


All passed
PS C:\SJSURepo\Data260\Data260-2837\code> python -m web_application.verify_hw07


```
{
  "homework": "HW7",
  "sid4": 2837,
  "commit_hash": "eb3e6d712d5e0cdde321819c0c2f17e5fdaa04b6",
  "model": "qwen2.5:1.5b (local Ollama)",
  "seed": 2837,
```

```
"verify_seed": 262837,
"generated_at_utc": "2026-10-26T21:26:01+00:00",
"checks": [
 {
  "check": "backend_running",
  "status": "PASS"
 },
 {
  "check": "login_valid_returns_token",
  "status": "PASS"
 },
 {
  "check": "login_invalid_returns_401",
  "status": "PASS"
 },
 {
  "check": "auth_me_without_token_401",
  "status": "PASS"
 },
 {
  "check": "auth_me_returns_identity_and_role",
  "status": "PASS"
 },
 {
  "check": "reader_can_list_200",
  "status": "PASS"
 },
 {
  "check": "reader_create_forbidden_403",
  "status": "PASS"
 },
 {
  "check": "writer_create_201",
  "status": "PASS"
 },
 {
  "check": "writer_delete_forbidden_403",
  "status": "PASS"
 },
 {
  "check": "moderator_delete_succeeds",
  "status": "PASS"
 },
 {
  "check": "agent_loop_completes_verification_task",
  "status": "PASS"
 },
 {
```

```
      "check": "safety_gate_blocks_destructive_commands",
      "status": "PASS"
    }
  ],
  "summary": "12/12 passed"
}
PS C:\SJSURepo\Data260\Data260-2837\code>
```


AI_USE

1. What I used an AI assistant for and what I did myself. I used Claude to review my repository against the assignment, fix import errors, write and refine the agent loop, Reflexion, verifier, safety gate, file tools and run scripts, and draft this report. I ran MySQL, MongoDB and Ollama, applied the fixes, executed every run, took the screenshots and committed the code myself.

2. One output that was wrong or unsuitable. The first safety gate passed the 10 required commands but allowed commands such as “echo hi; rm -fr /tmp/x” and “cat a.txt >/etc/passwd”. The AI also wrongly called the required variable db_session_basede26 a typo.

3. How I detected it. I ran extra bypass commands beyond the required ten and saw them allowed, and I checked database.py and found the required variable name.

4. What I changed and why it works now. The gate now uses word-boundary deny patterns and refuses shell chaining or redirection unless confirmed; all required and extra commands are handled correctly. I kept the required db_session_basede26 name.